

2024 Ph H2 Q2

Section: Our Dynamic Universe

Topic: Forces

Summary of Question

A car (1650 kg with passengers) tows a caravan (1350 kg) up a 9.6° slope.

Tasks:

- (a)(i) Calculate component of total weight down the slope.
- (a)(ii) Find forward force required at constant speed (friction = 1800 N).
- (b)(i) Show acceleration is 0.022 m s^{-2} when speeding up to 9.5 m s^{-1} in 250 s.
- (b)(ii) Calculate forward force while accelerating.
- (b)(iii) State assumption.
- (c) On level road, find tension in coupling (friction: car 740 N, caravan 1200 N).

(a)(i) Component of weight down slope

Total mass = $1650 + 1350 = 3000 \text{ kg}$.

Weight = $mg = 3000 \times 9.8 = 29\,400 \text{ N}$.

Component down slope = $W \sin \theta = 29\,400 \times \sin(9.6^\circ)$.

= $29\,400 \times 0.167 = 4900 \text{ N}$.

Final answer: $4.9 \times 10^3 \text{ N}$

(a)(ii) Forward force at constant speed

At constant speed: forward force = downslope component + friction.
 $= 4900 + 1800 = 6700 \text{ N}.$

Final answer: $6.7 \times 10^3 \text{ N}$

(b)(i) Acceleration

Initial speed $u = 4.0 \text{ m s}^{-1}$, final speed $v = 9.5 \text{ m s}^{-1}$, time $t = 250 \text{ s}.$

$$a = (v - u)/t = (9.5 - 4.0)/250 = 5.5/250 = 0.022 \text{ m s}^{-2}.$$

Final answer: 0.022 m s^{-2}

(b)(ii) Forward force while accelerating

$$\text{Extra force} = ma = 3000 \times 0.022 = 66 \text{ N}.$$

Total forward force = downslope + friction + ma .

$$= 4900 + 1800 + 66 = 6766 \text{ N}.$$

Final answer: $6.77 \times 10^3 \text{ N}$

(b)(iii) Assumption

Air resistance is included within 'friction' and does not change during acceleration.

(c) Tension in coupling

Total mass = 3000 kg, acceleration = $0.16 \text{ m s}^{-2}.$

For caravan alone: $F = ma + \text{friction}.$

$$= (1350 \times 0.16) + 1200 = 216 + 1200 = 1416 \text{ N}.$$

Final answer: $1.42 \times 10^3 \text{ N}$

Revision Tips

Always resolve weight into components along the slope.

Constant speed $\rightarrow$ driving force balances downslope + friction.

Check acceleration carefully using $(v-u)/t$.

Coupling tension = force needed to pull caravan alone.